CHAPTER 1

Data Analysis with jamovi

A student club surveyed 100 classmates about sleep, screen time, breakfast, and a short quiz. The club wants to know what the answers show. This chapter introduces Jamovi, which is a free software that does the data analysis calculations for you.

1.1 Getting Started with jamovi

jamovi is a free statistics program. You choose an analysis from a menu, tick a few boxes, and the results appear right away.

You can use jamovi in two ways:

- In a web browser, at `https://cloud.jamovi.org`, with a free Guest account. Nothing needs to be installed.
- On your own computer, by downloading it from `https://www.jamovi.org`. This version works without the internet.

The jamovi window has three parts:

- On the left, the *spreadsheet* holds the data. Each row is one person or thing, and each column is one variable.
- Across the top, the *analysis ribbon* lists the analyses you can run.
- The *results panel* on the right shows tables and graphs.

To try it, open an example dataset. Click the menu icon (three lines) in the top-left corner, then choose Open, Data Library, Examples, and Tooth Growth. On the Analyses tab, click Exploration, then Descriptives, and move `len` into the Variables box. A table of statistics appears without pressing any Run button.

Exercise 1 Which Part of the Window?

Name the part of the window you would use for each task: spreadsheet, analysis ribbon, or results panel.

Working Space

Task	Part
Fix a number that was typed wrong	
Choose an analysis	
Read the mean	

Answer on Page 7

1.2 Setting Up Your Data

Find `student_survey.csv` in your digital resources. Click the menu icon, choose Open, then Browse, and select the file. In jamovi Cloud, you may need to upload it first.

Column	What it records
`student_id`	A code number for each student
`travel`	How the student gets to school: Walk, Cycle, Bus, or Car
`breakfast`	Whether the student ate breakfast: Yes or No
`screen_time`	Hours on screens per day
`sleep_hours`	Hours of sleep per night
`sleep_quality`	The student's sleep rating: Poor, Fair, or Good
`quiz_score`	Score on a 20-point quiz

jamovi gives each column a *measure type*, which tells it what kind of data the column holds:

- Nominal: categories with no order, such as travel method
- Ordinal: categories with an order, such as Poor, Fair, Good
- Continuous: measured numbers, such as hours of sleep
- ID: labels for each case, such as a student code

jamovi guesses these types when it opens a file, and it can guess wrong. To fix one, double-click the column name and pick the correct type.

Two cells in `sleep_hours` are empty because those students skipped the question. jamovi treats empty cells as *missing values* and leaves them out of calculations.

Exercise 2 Measure Types

Give the correct measure type for each column.

Working Space

Column	Measure type
`student_id`	
`breakfast`	
`sleep_hours`	
`sleep_quality`	

Answer on Page 7

1.3 Describing Data

Click Exploration, then Descriptives. Move `quiz_score` and `sleep_hours` into the Variables box.

	`quiz_score`	`sleep_hours`
N	100	98
Missing	0	2
Mean	12.40	7.51
Median	12	7.60
Standard deviation	3.35	0.99
Minimum	4	4.5
Maximum	20	9.8

The average quiz score was 12.4 out of 20. Students slept 7.51 hours a night on average.

To compare groups, move a category into the Split by box. To draw a graph, open Plots and tick Histogram or Box plot. For a category such as `travel`, tick Frequency tables to

count how many students are in each group.

Exercise 3 Breakfast and Quiz Scores

Put `quiz_score` in the Variables box and `breakfast` in the Split by box.

1. What is the median quiz score for each group?
2. Which group did better?

Working Space

Answer on Page 7

1.4 Comparing Two Groups

Students who ate breakfast averaged 13.28 on the quiz, and students who skipped it averaged 11.03. A t-test checks whether a gap this large could be due to chance.

Click T-Tests, then Independent Samples T-Test. Move `quiz_score` into Dependent Variables and `breakfast` into Grouping Variable. Under Tests, tick Welch's and untick Student's. Welch's version is the safer choice because it does not require the two groups to have the same spread.

jamovi reports $t = -3.40$ and $p = 0.001$. A small p-value, below 0.05, means the gap is unlikely to be due to chance alone.

The negative sign appears because jamovi subtracts the Yes group from the No group. The breakfast group still scored higher.

These data come from a survey, so they show a link between breakfast and quiz scores. They cannot show that breakfast causes higher scores.

Exercise 4 Reading the t-Test

Working Space

1. Is $p = 0.001$ below 0.05?
2. Write one sentence explaining what the test shows.

Answer on Page 7

1.5 Looking at Relationships

Does more screen time go with less sleep? Click Exploration, then Scatterplot. Put `screen_time` on the X-Axis and `sleep_hours` on the Y-Axis.

The points slope downward: more screen time goes with less sleep.

Click Regression, then Correlation Matrix, and add both variables. The *correlation* r runs from -1 to 1 and measures how closely the points follow a straight line. Here $r = -0.70$, a strong negative relationship.

For a line of best fit, open Regression, then Linear Regression. Put `sleep_hours` in Dependent Variable and `screen_time` in Covariates. jamovi gives the line

$$\widehat{\text{sleep}} = 9.55 - 0.44 \times \text{screen time}$$

Each extra hour of screen time goes with about 0.44 fewer hours of sleep.

Exercise 5 Making a Prediction

Use the line to predict the sleep of a student with 6 hours of screen time.

Working Space

Answer on Page 8

1.6 Saving Your Work

Click the menu icon and choose Save As. jamovi saves an *.omv file*, which keeps your data and results together. In jamovi Cloud, download a copy to your own device before closing the tab.

To put a table or graph in a report, right-click it and choose Copy, then paste it into your document.

Exercise 6 Saving Choices

Working Space

1. Which file type keeps both your data and your results?
2. How do you move a table into a report?

Answer on Page 8

With jamovi you can open data, set it up, describe it, test it, and save your results, all without writing code. Because it is free and runs in a browser, you can use it on almost any device.

Appendix A

Answers to Exercises

Answer to Exercise 1 (on page 2)

Task	Part
Fix a number that was typed wrong	Spreadsheet
Choose an analysis	Analysis ribbon
Read the mean	Results panel

Answer to Exercise 2 (on page 3)

Column	Measure type
`student_id`	ID
`breakfast`	Nominal
`sleep_hours`	Continuous
`sleep_quality`	Ordinal

Answer to Exercise 3 (on page 4)

1. Breakfast: 13 points. No breakfast: 11 points.
2. Students who ate breakfast had the higher median score.

Answer to Exercise 4 (on page 5)

1. Yes.
2. Students who ate breakfast scored higher on the quiz, and the difference is too large to blame on chance alone.

Answer to Exercise 5 (on page 5)

$9.55 - 0.44 \times 6 = 6.91$ hours of sleep.

Answer to Exercise 6 (on page 6)

1. An .omv file.
2. Right-click the table, choose Copy, and paste it into the report.

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